

The Linux Terminal Command Reference

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A complete, task-organized reference to the terminal commands available on a standard GNU/Linux system. Every entry gives the command, what it does, and a typical example you can adapt.

 **Scope note.** Linux has no fixed, finite list of commands: any executable file found through the PATH environment variable is a command, so the total differs from machine to machine. This document covers (1) the standard command set shipped with mainstream distributions — GNU coreutils, util-linux, Bash built-ins, iproute2, systemd and common administration tools — and (2) widely installed third-party tools, clearly marked. The final section shows how to enumerate every command that exists on your own machine. Flags follow GNU conventions; BusyBox, BSD and macOS variants may differ.

1. How to Read This Reference

Commands are grouped by the task they perform. Find your task in the table of contents below, then scan the Command / Purpose / Example table in that section.

- Text in **bold monospace-style names** is a command you type; arguments in italics are placeholders you replace.
- Most commands accept **--help** for a full option list and have a manual page (see Section 2).
- Commands marked **[root]** normally require administrative privileges; run them through **sudo**.
- Commands marked **[pkg]** are not installed by default on every distribution and come from an additional package.

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2. Getting Help and Discovering Commands

Command	Purpose	Example
man	Show the full manual page for a command, file or function; press q to quit.	man chmod
man -k / apropos	Search the short descriptions of all manual pages by keyword.	apropos "list directory"
whatis		whatis rsync

	Print a one-line summary of a command from its manual page.	
info	Open the GNU Info hypertext documentation (often more complete than man for GNU tools).	info sed
command --help	Print a quick usage summary and option list directly from the command.	find --help
which	Show the path of the executable that would run for a command name.	which python3
whereis	Locate the binary, source files and manual pages of a command.	whereis docker
type	Show how the shell interprets a name: builtin, alias, keyword or external file.	type -a echo
alias	Define or list command shortcuts (aliases).	alias ll='ls -aF'
unalias	Remove an alias.	unalias ll
help	Show help for Bash built-in commands.	help printf
history	List previously executed commands from shell history.	history 50
compgen	Generate all command, alias, builtin or function completions (Bash).	compgen -c sort -u

Manual pages are divided into sections (1 user commands, 2 system calls, 3 library functions, 4 devices, 5 configuration files, 6 games, 7 miscellany, 8 administration). Request a section explicitly when names collide: **man 5 passwd** opens the file format page instead of the command page.

3. Shell Syntax Essentials

These metacharacters work in Bash and most POSIX shells. They combine and redirect commands and are used constantly in the examples throughout this document.

3.1 Redirection, Pipes and Operators

Syntax	Meaning
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	Pipe: send the standard output of the left command into the standard input of the right command.
>	Redirect standard output to a file, overwriting it.
>>	Append standard output to a file.
2>	Redirect standard error (file descriptor 2) to a file.
&>	Redirect both standard output and standard error (Bash).
2>&1	Send standard error to wherever standard output currently goes.
<	Redirect a file into the command's standard input.
<<EOF	Here-document: feed multiple lines of inline text as standard input until a line containing EOF.
<<<	Here-string: feed a single string as standard input.
;	Run commands sequentially regardless of success.
&&	Run the right command only if the left one succeeds (exit status 0).
	Run the right command only if the left one fails.
&	Run a command in the background.
()	Run commands in a subshell.
{ }	Group commands in the current shell (note the required spaces).
#	Start a comment to the end of the line.
\	Escape the next character so it is treated literally.

3.2 Globbing, Expansion and Quoting

Syntax	Meaning
*	Match any string of characters (does not match a leading dot by default).
?	Match exactly one character.
[abc] / [a-z]	Match one character from the set or range; named classes such as [:digit:] are supported.
{a,b,c}	Brace expansion: expand to each listed word; ranges use {1..5} or {a..z}.

~	The current user's home directory; ~user is that user's home.
\$VAR / \${VAR}	Expand a variable; braces delimit it from adjacent text.
\$()	Command substitution: replaced by the output of the inner command.
\$(())	Arithmetic expansion, e.g. echo $$(2 + 3)$.
<() / >()	Process substitution: present a command's input or output as a temporary file path.
'...'	Single quotes: everything inside is literal, no expansion occurs.
"..."	Double quotes: variables and command substitution still expand; globbing does not.

Combining the building blocks

```

1  # Count the .log files in /var/log
2  ls /var/log/*.log | wc -l
3
4  # Save both stdout and stderr, append, and keep running on failure
5  make >> build.log 2>&1 || echo "build failed"
6
7  # Loop over brace-expanded names
8  mkdir -p project/{src,docs,tests}
9
10 # Feed inline text to a command
11 sort <<EOF
12 banana
13 apple
14 cherry
15 EOF

```

4. Files and Directories

Command	Purpose	Example
pwd	Print the current working directory.	pwd
ls	List directory contents. Common flags: -l long format, -a include hidden files, -h human-readable sizes, -t sort by time, -S by size, -R recursively, -i show inode, -d list the directory itself.	ls -lah /etc

cd	Change directory. cd alone goes home; cd - returns to the previous directory; cd .. moves up one level.	cd /var/log
tree [pkg]	Show a directory tree with levels.	tree -L 2 src
mkdir	Create directories; -p creates parents and does not error if they exist, -m sets the mode.	mkdir -p a/b/c
rmdir	Remove empty directories only.	rmdir old_dir
rm	Remove files or directories: -r recursive, -f force without prompting, -i ask each time, -v verbose. There is no trash can; deletion is final.	rm -iv file.txt
cp	Copy files or directories: -r/-R recursive, -a archive (preserve permissions and timestamps), -n no-clobber, -u update only, -i interactive.	cp -av src/ backup/
mv	Move or rename files and directories; -i prompts before overwrite, -n never overwrites.	mv old.txt new.txt
touch	Create an empty file or update a file's access and modification timestamps.	touch notes.txt
ln	Create links: -s makes a symbolic link; without -s it makes a hard link.	ln -s /opt/app/bin/app ~/bin/app
stat	Show detailed metadata: size, inode, permissions, owner and timestamps.	stat /etc/hosts
file	Detect a file's actual type regardless of its name.	file image.png
basename	Strip the directory portion of a path; optionally strip a suffix.	basename /a/b/c.txt .txt
dirname	Print only the directory portion of a path.	dirname /a/b/c.txt
realpath	Resolve the absolute, symlink-free path.	realpath ./script.sh
readlink	Print the target of a symbolic link; -f resolves recursively.	readlink -f /usr/bin/python3
mktemp	Create a temporary file or directory with a unique name.	mktemp -d
rename		rename 's/\.jpeg\$/\.jpg/' *.jpeg

	Rename multiple files using a Perl-style expression (util-linux version).	
install	Copy files while setting ownership and permissions in one step (used by build systems).	<code>install -m 755 script.sh /usr/local/bin/</code>
truncate	Shrink or extend a file to a given size; -s 0 empties it.	<code>truncate -s 0 big.log</code>
shred	Overwrite a file repeatedly to make recovery harder; -u removes it afterwards.	<code>shred -u secret.key</code>

 **Destructive commands.** `rm -rf`, `dd`, `mkfs`, `fdisk` and `shred` can destroy data instantly and irreversibly. Always resolve relative paths, expand globs before deleting (run the equivalent `ls` first), and never run a destructive command as root until the target is verified.

5. Viewing File Contents

Command	Purpose	Example
cat	Concatenate and print files; -n numbers lines, -A shows control characters and line endings.	<code>cat -n config.ini</code>
tac	Print a file in reverse line order.	<code>tac logfile</code>
nl	Print a file with line numbers, with control over numbering style.	<code>nl -ba file</code>
less	Page through text interactively; supports searching with /, g/G for top/bottom, q to quit. Preferred pager for large files.	<code>less /var/log/syslog</code>
more	Older basic pager; less is almost always preferred.	<code>more file.txt</code>
head	Print the first lines; -n sets the count, -c bytes.	<code>head -n 20 file</code>
tail	Print the last lines; -f follows a growing file in real time, -F retries if the file is recreated.	<code>tail -F /var/log/nginx/access.log</code>
echo	Print text or expanded variables; -n omits the trailing newline, -e interprets escapes.	<code>echo "\$PATH"</code>

printf	Formatted output using a format string, like the C function.	<code>printf '%02d\n' 5</code>
strings	Extract printable text strings from a binary file.	<code>strings program grep -i version</code>
od	Dump a file in octal, hexadecimal or character form.	<code>od -c -Ax file.bin</code>
hexdump	Display bytes in hexadecimal and ASCII columns.	<code>hexdump -C file.bin</code>
xxd	Hex dump, or reverse one back into binary with <code>-r</code> .	<code>xxd file.bin</code>
fold	Wrap long lines at a fixed width.	<code>fold -w 80 file</code>
fmt	Reformat paragraphs to a target width.	<code>fmt -w 72 draft.txt</code>
rev	Reverse characters on each line.	<code>rev words.txt</code>
zcat / zless / zgrep	cat, less and grep that work directly on gzip-compressed files.	<code>zgrep ERROR app.log.gz</code>

5.1 Terminal Text Editors

Command	Purpose	Example
nano	Beginner-friendly modeless editor; shortcut prompts are shown on screen.	<code>nano ~/.bashrc</code>
vi / vim	Modal editor present on virtually every system; <code>i</code> enters insert mode, <code>Esc</code> returns to normal mode, <code>:wq</code> saves and quits, <code>:q!</code> quits without saving.	<code>vim /etc/hosts</code>
emacs [pkg]	Extensible editor and computing environment; terminal mode with <code>-nw</code> .	<code>emacs -nw notes.org</code>
view	Open vi/vim in read-only mode.	<code>view important.conf</code>

6. Text Processing and Filtering

These commands read text (usually through a pipe), transform it, and write the result to standard output. They are the core of command-line data work.

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Command	Purpose	Example
grep	Search text for lines matching a pattern. Key flags: -i ignore case, -r recursive, -v invert, -n line numbers, -E extended regex, -F fixed strings, -c count, -l filenames only, -o only the match, -A/-B/-C context lines.	grep -rn "TODO" src/
egrep / fgrep	Equivalent to grep -E (extended regex) and grep -F (literal text); modern grep prefers the explicit flags.	egrep 'error fail' log
sed	Stream editor: substitute, delete, insert or transform lines non-interactively. -i edits files in place.	sed -i 's/foo/bar/g' file
awk	Column-oriented programming language: select fields, compute, format reports. \$1, \$2 are fields, \$0 the whole line, -F sets the field separator.	awk -F: '{print \$1}' /etc/passwd
cut	Extract columns: -d delimiter with -f fields, or -c character ranges.	cut -d',' -f2 data.csv
sort	Sort lines: -n numeric, -h human-readable sizes, -r reverse, -k key column, -t field separator, -u unique, -M month names.	sort -t',' -k3 -n sales.csv
uniq	Collapse adjacent duplicate lines; -c counts, -d lists only duplicates, -u only uniques. Input is usually sorted first.	sort names uniq -c
wc	Count lines (-l), words (-w), characters (-m) and bytes (-c).	wc -l file
tr	Translate or delete characters: -d deletes a set, -s squeezes repeats; it reads standard input only.	tr 'a-z' 'A-Z' < file
tee	Send input both to a file and to standard output; -a appends.	make 2>&1 tee build.log
xargs	Turn standard input lines into arguments for another command; -0 pairs with find -print0.	find . -name '*.tmp' -print0 xargs -0 rm
paste	Merge corresponding lines of files side by side; -d sets the delimiter.	paste -d',' a.txt b.txt

join	Join two files on a shared field, like a database join.	join -1 2 -2 1 a.txt b.txt
comm	Compare two sorted files line by line in three columns: only-in-A, only-in-B, in-both.	comm -12 sorted1 sorted2
column	Format input into aligned columns or a table; -t auto-aligns, -s sets the input delimiter.	column -t -s',' data.csv
diff	Show line differences between files; -u produces unified context, -r compares directories.	diff -u old new > patch.diff
cmp	Byte-level comparison; reports the first differing byte.	cmp a.iso b.iso
patch	Apply a diff produced by diff -u to the original file.	patch -p1 < fix.diff
expand / unexpand	Convert tabs to spaces, or spaces back to tabs; -t sets tab stops.	expand -t 4 file
split	Split a file into pieces by lines (-l), size (-b) or on patterns.	split -b 100M big.bin part_
csplit	Split a file at context patterns into named pieces.	csplit log '/^==/' '{*}'
iconv	Convert text between character encodings; -l lists them.	iconv -f GBK -t UTF-8 in.txt
dos2unix / unix2dos [pkg]	Convert CRLF Windows line endings to LF and back.	dos2unix script.sh

7. Searching Files and Content

Command	Purpose	Example
find	Search a directory tree by name, type, size, time, owner or permissions and act on results. Tests: -name, -iname, -type f/d/l, -size, -mtime/-atime/-ctime, -user, -group, -perm, -maxdepth; actions: -print, -delete, -exec command {} +, -ok.	find . -type f -name '*.log' -mtime +30
locate		locate -i nginx.conf

	Find paths instantly from a prebuilt filename index.	
updatedb [root]	Rebuild the database used by locate.	sudo updatedb
grep -r	Recursive content search inside files; add -l to skip binaries, --include to filter filenames.	grep -rln --include='*.py' 'class' .
fd [pkg]	Modern, fast, user-friendly alternative to find.	fd -e pdf report
fzf [pkg]	Interactive fuzzy finder for any list, commonly used to pick files or history entries.	vim \$(fzf)

find patterns worth memorizing

```

1  # Files larger than 500 MB
2  find / -type f -size +500M 2>/dev/null
3
4  # Files modified in the last 24 hours under /etc
5  find /etc -type f -mtime -1
6
7  # Change ownership of every found file safely (NUL-delimited)
8  find . -type f -name '*.sh' -print0 | xargs -0 chmod +x
9
10 # Find and delete empty directories
11 find . -type d -empty -delete

```

8. Permissions, Ownership and File Attributes

Command	Purpose	Example
chmod	Change permission bits. Octal form (e.g. 755, 644) or symbolic form (u+x, g-w, o=r, a+rX). Special bits: setuid 4xxx, setgid 2xxx, sticky 1xxx; -R recurses.	chmod 640 file; chmod u+x script.sh
chown	Change owner and optionally group; -R recurses, --reference copies another file's ownership.	sudo chown alice:staff file

chgrp	Change group ownership only.	chgrp devteam project
umask	Show or set the default permission mask applied to newly created files.	umask 027
getfacl / setfacl [pkg]	Read and set POSIX access control lists for per-user or per-group permissions.	setfacl -m u:bob:rwX dir
lsattr / chattr	List or change extended filesystem attributes such as +i immutable and +a append-only.	sudo chattr +i /etc/resolv.conf
chcon / restorecon	Set or restore SELinux security contexts on SELinux-enabled systems.	sudo restorecon -v /var/www/html

The three octal digits encode **owner / group / others**; each digit is the sum of read = 4, write = 2 and execute = 1. For example 754 means the owner may read, write and execute (7); the group may read and execute (5); others may only read (4).

9. Users, Groups and Accounts

Command	Purpose	Example
whoami	Print the current effective username.	whoami
id	Print user ID, group ID and all group memberships.	id alice
groups	List the groups a user belongs to.	groups
who	List users currently logged in.	who
w	Show logged-in users and what they are currently running.	w
users	Print a compact list of logged-in usernames.	users
last / lastb	Show history of successful (last) or failed (lastb) logins.	last -n 10
lastlog	Report the most recent login of every account.	lastlog
passwd	Change a password; root can change any account and lock/unlock with -l/-u.	passwd; sudo passwd bob

chage	View and change password aging and expiry policy.	sudo chage -l alice
chsh / chfn	Change a user's login shell or finger/gecos information.	chsh -s /bin/bash
useradd [root]	Create a system user account; -m creates the home directory, -s sets the shell, -G adds supplementary groups. Many systems offer the friendlier adduser wrapper.	sudo useradd -m -s /bin/bash bob
userdel [root]	Delete an account; -r also removes the home directory and mail spool.	sudo userdel -r bob
usermod [root]	Modify an account: -aG append groups, -L/-U lock/unlock, -s shell, -d home.	sudo usermod -aG docker alice
groupadd / groupdel / groupmod [root]	Create, delete or rename/modify a group.	sudo groupadd devteam
gpsswd	Administer /etc/group: set a group password or add/remove members.	sudo gpsswd -a alice devteam
su	Switch user; - starts a login shell that loads the target's environment.	su - bob
sudo	Run a command with elevated privileges according to /etc/sudoers; -i opens a root login shell, -u runs as another user.	sudo systemctl restart nginx
sudoedit	Edit a file with elevated rights using your own editor.	sudoedit /etc/fstab
visudo [root]	Safely edit /etc/sudoers with syntax validation.	sudo visudo
newgrp / sg	Start a shell with a different primary group.	newgrp devteam

10. Process and Job Management

Command	Purpose	Example
ps	Snapshot current processes. ps aux (BSD style) and ps -ef (POSIX style) show every process with user, PID and command.	ps aux grep nginx

ps	Show processes as a parent-child tree; -p adds PIDs.	ps -p
pidof	Print the PID(s) of a running program by name.	pidof sshd
pgrep	Find PIDs matching a name or pattern; -a shows the command line, -u filters by user, -f matches the full command line.	pgrep -a -u alice
top	Live, interactive process monitor sorted by CPU; press P for CPU, M for memory, k to kill, q to quit.	top
htop [pkg]	Enhanced interactive process viewer with scrolling and mouse support.	htop
kill	Send a signal to a PID. Default is TERM (15); kill -9 sends KILL which cannot be ignored; -l lists signals.	kill -15 1234
killall	Signal processes by name rather than PID.	killall -HUP nginx
pkill	Signal processes matching a pattern; uses the same selectors as pgrep.	pkill -f backup.sh
jobs	List background and stopped jobs of the current shell with job numbers.	jobs -l
bg / fg	Resume a stopped job in the background, or bring a job to the foreground.	fg %1
nohup	Run a command immune to hangup so it survives logout; output defaults to nohup.out.	nohup ./longjob &
disown	Detach a running background job from the shell so it is not sent SIGHUP on exit.	disown %1
nice / renice	Start a process with a niceness value, or change the priority (and with -p a PID) of a running one.	renice +10 -p 1234
time	Measure elapsed, user and system time of a command.	time ./build.sh
timeout	Run a command and kill it if it exceeds a time limit; -k sends a follow-up KILL.	timeout 30s curl URL
watch	Re-run a command on an interval and show its output full screen; -n sets seconds, -d	watch -n 2 df -h

	highlights changes.	
lsuf	List open files, including the processes using network ports (-i) or a specific file.	sudo lsuf -i :22
fuser	Identify or kill processes using a file, directory or socket; -k kills them.	fuser -v /mnt/usb
strace [pkg]	Trace system calls and signals of a process for debugging; -f follows children, -p attaches.	strace -p 1234
ltrace [pkg]	Trace library calls of a process.	ltrace ./program
taskset	Pin a process to specific CPU cores; -pc views/affinity-changes by PID.	taskset -pc 0,1 1234

11. System and Hardware Information

Command	Purpose	Example
uname	Print kernel and machine information; -a prints everything, -r the kernel release, -m the architecture.	uname -a
hostname	Show or set the system hostname; -l lists assigned IP addresses.	hostname -l
hostnamectl	Query or permanently set the hostname and related machine settings (systemd).	hostnamectl set-hostname web01
uptime	Show running time, number of users and the 1/5/15-minute load averages.	uptime
dmesg	Print the kernel ring buffer (boot and hardware messages); -w follows new messages, -T human times.	sudo dmesg -T tail
lscpu	Show CPU architecture, core/thread counts, model and flags.	lscpu
nproc	Print the number of available processing units.	nproc
lsblk	List block devices (disks and partitions) as a tree with sizes and mount points; -f adds filesystem/UUID info.	lsblk -f

lspci	List PCI buses and attached devices; -k shows the driver in use.	lspci -k
lsusb	List USB buses and connected devices; -t shows the tree, -v full detail.	lsusb -t
lshw	Detailed hardware inventory; -short gives a compact summary.	sudo lshw -short
dmidecode [root]	Read SMBIOS/DMI tables: BIOS, motherboard, memory slots and serial numbers.	sudo dmidecode -t memory
free	Show memory and swap usage; -h uses human-readable units, -s samples repeatedly.	free -h
vmstat	Report virtual memory, run queue, paging and CPU statistics over intervals.	vmstat 2 5
mpstat	Per-CPU utilization statistics (sysstat package).	mpstat -P ALL 1
iostat	CPU and per-device disk I/O statistics; -x adds extended metrics.	iostat -xz 1
sar	Collect and replay historical system activity reports.	sar -u 1 5
lsmod / modinfo	List loaded kernel modules, or show a module's details.	lsmod grep kvm
modprobe / insmod / rmmod [root]	Load (with dependencies) or unload kernel modules.	sudo modprobe vfat
sysctl	Read or change kernel parameters at runtime; -a lists all, -w sets, permanent entries live in /etc/sysctl.conf.	sysctl vm.swappiness
date	Print or set the date and time with flexible formatting; supports date arithmetic.	date '+%Y-%m-%d %H:%M'
timedatectl	Query and configure time, timezone and NTP synchronization (systemd).	timedatectl set-timezone Asia/Shanghai
hwclock	Read or set the hardware (RTC) clock and sync it with the system clock.	sudo hwclock --systohc
cal	Display a calendar; -y shows the whole year.	cal 12 2026

locale / localectl	Show locale settings, or set system locale and keyboard layout.	localectl status
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Distribution identity is stored in **/etc/os-release**; display it with **cat /etc/os-release**. The live kernel exposes tunable state under the **/proc** filesystem (for example **/proc/cpuinfo**, **/proc/meminfo**, **/proc/loadavg**).

12. Disks, Filesystems and Mounts

Command	Purpose	Example
df	Report filesystem free and used space; -h human-readable, -T shows filesystem type, -i reports inodes.	df -hT
du	Estimate disk usage of files and directories; -sh summarizes a directory, -h human units, --max-depth limits levels, -a all files.	du -sh --max-depth=1 /var
ncdu [pkg]	Interactive disk-usage browser for finding space hogs.	ncdu /home
blkid	Print UUID, label and filesystem type of block devices.	sudo blkid
fdisk [root]	Partition disks with an interactive menu; -l lists partition tables.	sudo fdisk -l /dev/sda
parted [root]	Scriptable partition editor supporting GPT and large disks.	sudo parted /dev/sda print
mkfs [root]	Build a filesystem; type-specific variants include mkfs.ext4, mkfs.xfs, mkfs.btrfs, mkfs.vfat, mkfs.ntfs.	sudo mkfs.ext4 /dev/sdb1
mkswap / swapon / swapoff [root]	Format a swap area and enable or disable it.	sudo swapon --show
fsck / e2fsck [root]	Check and repair a filesystem; always run it on an unmounted device.	sudo fsck -f /dev/sdb1
tune2fs	Tune ext2/3/4 parameters such as labels, reserved blocks and check intervals.	sudo tune2fs -L data /dev/sdb1

resize2fs	Expand or shrink an ext filesystem to match its resized partition.	<code>sudo resize2fs /dev/sda2</code>
mount / umount [root]	Attach a filesystem to a directory tree, or detach it; mount alone lists active mounts. Common options: -t type, -o ro/rw/noexec/noatime.	<code>sudo mount /dev/sdb1 /mnt</code>
findmnt	Display mounted filesystems as a tree or verify a target.	<code>findmnt /mnt</code>
dd	Low-level byte copy and conversion between files or devices. Extremely powerful and dangerous: if and of point at whole disks, data is overwritten with no confirmation.	<code>dd if=image.iso of=/dev/sdb bs=4M status=progress</code>
wipefs	Erase filesystem signatures from a device.	<code>sudo wipefs -a /dev/sdb</code>
sync	Flush cached writes to persistent storage; run before unplugging media.	<code>sync</code>
eject	Eject removable media such as a CD/DVD or USB device.	<code>eject /dev/sr0</code>
smartctl [pkg, root]	Read SMART health data and run disk self-tests (smartmontools).	<code>sudo smartctl -a /dev/sda</code>
hdparm	Query or tune SATA/ATA drive parameters.	<code>sudo hdparm -l /dev/sda</code>

LVM (Logical Volume Manager) adds **pvcreate/pvs**, **vgcreate/vgs**, **lvcreate/lvs/lvextend** for physical volumes, volume groups and logical volumes respectively.

13. Software Package Management

Package commands depend on the distribution family. Use the column that matches your system.

13.1 Debian, Ubuntu and Derivatives (APT / dpkg)

Command	Purpose
apt update	Refresh the local index of available packages [root].

apt upgrade	Upgrade all upgradable packages; full-upgrade may install/remove dependencies.
apt install package	Install a package and its dependencies.
apt remove / purge	Remove a package; purge also deletes its configuration files.
apt autoremove	Remove packages installed only as dependencies that are no longer needed.
apt search / show	Search package descriptions or show full package details.
apt list --installed	List installed packages.
dpkg -i file.deb	Install a local .deb file directly.
dpkg -l / -L / -S	List installed packages; list files owned by a package; find which package owns a file.

13.2 Red Hat, Fedora, CentOS, Rocky and openSUSE (DNF / YUM / RPM)

Command	Purpose
dnf check-update	Refresh metadata and list available updates.
dnf update	Apply all updates.
dnf install / remove	Install or erase a package.
dnf search / info	Search repositories or show package details.
dnf autoremove	Remove unneeded dependency packages.
dnf provides path	Find which package supplies a given file or command.
rpm -ivh file.rpm	Install a local RPM file verbosely with hash progress.
rpm -qa / -ql / -qf	List all installed packages; list a package's files; identify the owner of a file.
zypper	The openSUSE/SLES equivalent: zypper refresh, install, remove, search, update.

13.3 Arch Linux, Alpine and Cross-Distribution Systems

Command	Purpose
pacman -Syu	Arch: synchronize databases and fully upgrade the system.
pacman -S / -Rns / -Ss / -Q	Arch: install; remove with dependencies and config; search repositories; query installed packages.
apk add / del / search / update / upgrade	Alpine Linux package management.
snap install / list / refresh / remove	Manage Canonical Snap packages.
flatpak install / list / update / uninstall	Manage sandboxed Flatpak applications.
make && sudo make install	Build and install software from source after ./configure or cmake; prefer native packages when available.

14. Networking and the Internet

Command	Purpose	Example
ip	Modern network configuration tool (iproute2) replacing ifconfig and route. ip addr shows addresses, ip link manages interfaces, ip route manages routes, ip neigh shows the ARP table.	ip -br addr
ifconfig / route / arp	Legacy net-tools; deprecated in favor of ip but still seen on older systems.	ifconfig -a
nmcli	Command-line control of NetworkManager: devices, connections and Wi-Fi.	nmcli device status
ping / ping6	Send ICMP echo requests to test reachability and latency; -c limits count.	ping -c 4 example.com
traceroute / tracepath	Show the network path (hops) to a host.	tracepath example.com
mtr [pkg]	Combines ping and traceroute into a live per-hop quality report.	mtr -rwc 50 example.com
dig		dig MX example.com +short

	Query DNS records in detail; +short gives a terse answer, @server picks a resolver, ANY/A/AAAA/MX/TXT/CNAME select record types.	
nslookup	Interactive or one-shot DNS lookup, older than dig.	nslookup example.com 8.8.8.8
host	Simple DNS lookup utility.	host -t TXT example.com
whois [pkg]	Query domain registration and ownership records.	whois example.com
ss	List sockets, replacing netstat: -t TCP, -u UDP, -l listening, -p processes, -n numeric, -a all.	ss -tulpn
netstat	Legacy socket and routing table tool; use ss instead on modern systems.	netstat -tulpn
curl	Transfer data over HTTP/S, FTP and many protocols; -I headers, -L follow redirects, -O keep remote filename, -o output file, -X method, -d data, -H header, -k skip certificate validation.	curl -L -O https://site/file.tar.gz
wget	Non-interactive downloader; -c resumes, -r mirrors recursively, -q quiet, -P sets the output directory.	wget -c https://site/file.iso
nc / ncat	Netcat: raw TCP/UDP connections and listeners for port tests and data transfer; -l listens, -z scans, -v verbose.	nc -zv host 80
telnet	Unencrypted text protocol, now mostly used to test plain-text TCP ports.	telnet host 25
ftp / lftp	FTP client; lftp is the scriptable, feature-rich alternative.	lftp ftp://host
tcpdump [pkg, root]	Capture and display network packets; -i selects the interface, -w writes a pcap, expressions filter traffic.	sudo tcpdump -i any port 53
ethtool	Show or change NIC settings: link state, speed, duplex and offloads.	sudo ethtool eth0

iptables / nft [root]	Administer the kernel packet filter; nftables (nft) is the modern successor. -L lists rules, -F flushes.	sudo iptables -L -n -v
ufw	Simplified front-end firewall for Debian/Ubuntu: allow, deny, status, enable.	sudo ufw allow 22/tcp
firewall-cmd	firewalld front-end used on RHEL/Fedora: --add-port, --add-service, --permanent.	sudo firewall-cmd --list-all
getent	Look up entries from name-service databases (hosts, passwd, group, services) via /etc/nsswitch.conf.	getent hosts example.com

15. Secure Shell and Remote File Transfer

Command	Purpose	Example
ssh	Log in or run a command on a remote host over an encrypted channel; -p sets the port, -i selects a key, -L/-D/-R build tunnels.	ssh -p 22 alice@host
scp	Copy files over SSH; -r copies directories, -P sets the port (legacy; rsync is preferred for large transfers).	scp -r dir alice@host:~/
sftp	Interactive FTP-like file transfer over an SSH connection.	sftp alice@host
rsync	Fast incremental file synchronization locally or over SSH; -a archive, -v verbose, -z compress, --delete mirror deletions, -n dry run, -P show progress.	rsync -avzP src/ host:/dest/
ssh-keygen	Generate, manage and convert authentication key pairs; -t selects the algorithm, -b the key size.	ssh-keygen -t ed25519
ssh-copy-id	Install your public key into a remote account's authorized_keys.	ssh-copy-id alice@host

ssh-agent / ssh-add	Hold decrypted private keys in memory so passphrases are entered once; -l lists loaded keys.	eval \$(ssh-agent); ssh-add
ssh-keyscan	Collect remote host public keys for known_hosts bootstrapping.	ssh-keyscan host >> ~/.ssh/known_hosts

16. Archives and Compression

Command	Purpose	Example
tar	Pack and unpack tape archives. Flags: c create, x extract, t list, f file, z gzip, j bzip2, J xz, --zstd Zstandard, v verbose, -C change to directory.	tar czf out.tar.gz dir/
gzip / gunzip / zcat	Compress or expand .gz files; -k keeps the original, -9 maximum compression, -1 fastest.	gzip -9 data.csv
bzip2 / bunzip2	Higher-ratio .bz2 compression and decompression.	bzip2 file.txt
xz / unxz / xzcat	High-ratio LZMA2 .xz compression.	xz -T0 archive.tar
zstd / unzstd [pkg]	Fast modern Zstandard compression; -# sets the level, --train and -T0 threads are advanced options.	zstd -19 file
zip / unzip	Create and extract cross-platform .zip archives; -r recurses, -e encrypts, -l lists contents.	zip -r out.zip dir/
7z [pkg]	7-Zip: handles 7z, zip, tar and many other formats; a adds, x extracts, l lists.	7z x archive.7z
uncompress / unrar [pkg]	Handle legacy .Z files and RAR archives respectively.	unrar x file.rar

tar cheat sheet

```

1  tar czf archive.tar.gz dir/  # create gzip-compressed archive
2  tar cjf archive.tar.bz2 dir/ # create bzip2-compressed archive
3  tar cJf archive.tar.xz dir/  # create xz-compressed archive
4  tar tzf archive.tar.gz       # list contents without extracting

```

```
5 tar xzf archive.tar.gz # extract gzip archive
6 tar xf archive.tar.* -C /tmp # extract anything, into /tmp
```

17. Services, Scheduling and Logs

17.1 systemd Service Management: systemctl

Command	Purpose
systemctl start / stop name	Start or stop a unit now.
systemctl restart / reload name	Restart a unit, or reload its configuration without stopping it.
systemctl enable / disable name	Control whether a unit starts at boot; --now also starts/stops it immediately.
systemctl status / is-active / is-enabled name	Inspect runtime state, active state or boot state.
systemctl list-units / list-unit-files	List active units, or all installed unit files and their enablement.
systemctl daemon-reload	Reload unit files after editing a .service definition.
systemctl get-default / set-default target	Show or change the boot target (graphical.target, multi-user.target).
service name action	Legacy SysV-compatible wrapper still found on older systems.

17.2 Logs: journalctl and Traditional Logs

Command	Purpose	Example
journalctl	Query the systemd journal: -u unit, -f follow, -b this boot, --since/--until time window, -p priority, -k kernel messages, -n last lines, --disk-usage size.	journalctl -u nginx -f
logger	Write a message into syslog/the journal from scripts.	logger -t backup "job started"

logrotate	Rotate, compress and prune log files according to /etc/logrotate.d policy.	sudo logrotate -f /etc/logrotate.conf
------------------	--	---------------------------------------

17.3 Scheduling: cron and at

Command	Purpose	Example
crontab -e / -l / -r	Edit, list or remove the current user's scheduled jobs; system-wide jobs live in /etc/crontab and /etc/cron.d.	crontab -l
at / atq / atrm	Schedule a one-off command at a time, queue it, list the queue, or remove a queued job.	echo 'backup.sh' at 02:00
batch	Run a queued job when system load drops below a threshold.	batch < job.sh
anacron	Run periodic jobs that were missed while a machine was powered off.	anacron -f

A crontab line has five time fields — **minute (0-59)**, **hour (0-23)**, **day of month (1-31)**, **month (1-12)**, **day of week (0-7)** — followed by the command. ***** means every unit, ***/5** means every five, and lists/ranges such as **1,15** or **Mon-Fri** are accepted. Example: **30 2 * * 1 /usr/local/bin/weekly.sh** runs every Monday at 02:30.

18. Bash Built-in Commands

Built-ins run inside the shell itself rather than as external programs. Run **help** to list them all and **help name** for the syntax of one. The most frequently used ones:

Built-in	Purpose
alias / unalias	Define or remove command aliases.
bg / fg / jobs	Job control: background, foreground and list jobs.
bind	Display and modify Readline key bindings.
builtin / command	Force execution of a builtin, or bypass aliases/functions to run an external command.
caller	Return the context of a calling function or sourced script (debugging).

cd / pushd / popd / dirs	Change directory, or maintain a navigable directory stack.
compgen / complete / compopt	Programmable completion machinery.
declare / typeset / local / readonly	Declare variables and attributes (-i integer, -a indexed array, -A associative array, -r read-only, -x exported); local is used inside functions.
echo / printf	Output text; printf provides format strings.
enable	Enable or disable shell built-ins.
eval	Execute an assembled string as a shell command (use with caution).
exec	Replace the shell with a command, or redirect the current shell's file descriptors.
exit / logout	Leave the shell with an exit status, or log out of a login shell.
export / unset	Mark variables for child processes, or remove them.
fc	Recall and edit commands from history.
getopts	Parse positional options in scripts.
hash	Remember or display the resolved paths of commands.
help	Documentation for built-ins.
history	Access the command history.
kill	Send signals to jobs or PIDs.
let / (())	Integer arithmetic.
mapfile / readarray	Read input lines into an array.
read	Read a line of input into variables; -r keeps backslashes literal, -p prompts.
return / source (.)	Return from a function, or execute a script in the current shell.
set / shopt	Set shell options and positional parameters (set -e exit on error, -u unset-variable error, -x trace, -o pipefail); shopt toggles Bash-specific behavior.
shift	Drop positional parameters (\$2 becomes \$1).
test / [/ [[]]	Evaluate conditions: file tests, string tests and numeric comparisons; [[]] is the safer Bash form.
trap	Run cleanup code on signals or EXIT.

type	Explain how a name is resolved.
ulimit	Show or set per-process resource limits (-a all).
umask	Set the default creation permission mask.
wait	Block until background jobs finish and collect their statuses.
: / true / false	No-op that succeeds; true always returns 0, false always returns 1.
break / continue	Loop control inside for/while/until.

19. Environment and Terminal Control

Command	Purpose	Example
env	Run a command with a modified environment, or print all variables with no arguments.	env -i PATH=/bin bash
printenv	Print the value of one or all environment variables.	printenv HOME SHELL
export	Export variables to child processes; -p lists exports.	export EDITOR=vim
set	With no arguments, list every shell variable and function.	set grep -i path
tty	Print the device path of the current terminal.	tty
stty	Change terminal line settings; -a prints all current settings.	stty -a
tput	Query terminal capabilities: colors, cursor movement and text attributes.	tput cols; tput setaf 2
clear	Clear the terminal screen (same as Ctrl+L).	clear
reset	Reinitialize the terminal after binary output garbles it.	reset
script	Record an entire terminal session to a typescript file.	script session.log
tmux [pkg]	Terminal multiplexer: persistent sessions, split panes and windows; new, attach, detach (Ctrl+B then D), ls.	tmux new -s work

screen [pkg]	Older terminal multiplexer with detach/resume capability.	screen -S work
wall	Broadcast a message to every logged-in user's terminal [root].	wall "reboot in 10 minutes"
write	Send a one-to-one message to another logged-in user's terminal.	write alice pts/0
mesg	Control whether others can write to your terminal.	mesg n

20. Printing

Command	Purpose	Example
lp / lpr	Submit a file to a print queue; -d/-P selects the printer, -n sets copies.	lp -d office doc.pdf
lpq / lpstat	Show queued jobs, or printer/daemon status (-p printers, -o jobs, -t everything).	lpstat -t
lprm / cancel	Remove a queued print job by ID.	cancel 12
lpadmin	Add, remove or configure CUPS printers and the default printer.	lpadmin -d office

21. Power and Session Control

Command	Purpose	Example
shutdown [root]	Schedule a power-off or reboot with a warning to users; -r reboots, -h halts, now acts immediately, a time argument schedules it, -c cancels.	sudo shutdown -r +10 "maintenance"
reboot / poweroff / halt	Immediate reboot, power-off or halt.	sudo reboot
systemctl suspend / hibernate	Suspend to RAM or hibernate to disk.	systemctl suspend
init / telinit		sudo telinit 3

	Legacy runlevel/target switching; on systemd, telinit 3 changes the current target.	
loginctl	Inspect and control user sessions and seats; list-sessions, terminate-user, lock-session.	loginctl list-sessions
exit / logout	End the current shell or login session.	exit

22. Checksums, Encoding, Math and Miscellaneous Utilities

Command	Purpose	Example
md5sum / sha1sum / sha256sum / sha512sum	Compute or verify (-c) cryptographic checksums of files.	sha256sum -c SHA256SUMS
base64	Encode to or decode (-d) Base64.	base64 file; base64 -d out
bc	Arbitrary-precision calculator language; -l loads the math library.	echo 'scale=4; 22/7' bc -l
dc	Reverse-Polish-notation stack calculator.	echo '5 3 + p' dc
factor	Print the prime factors of numbers.	factor 360
units [pkg]	Convert between measurement units.	units '10 km' 'miles'
seq	Print a number sequence; supports start, step and end plus formatting.	seq -w 1 10
shuf	Randomly permute input lines or generate random picks; -n limits count, -i generates a range.	shuf -i 1-49 -n 6
yes	Repeatedly print a string (default y) to auto-answer interactive prompts.	yes ./installer
uuidgen	Generate a random UUID.	uuidgen
numfmt	Convert numbers between raw and human-readable magnitudes.	numfmt --to=iec 1048576

openssl	Swiss-army cryptography toolkit: certificates, hashes, ciphers, TLS clients.	openssl x509 -in cert.pem -text
getconf	Query system configuration limits (path length, page size, and similar).	getconf LONG_BIT
ldd	List the shared libraries a binary depends on.	ldd /bin/ls
ldconfig / ldd	Update or query the dynamic linker cache; -p prints known libraries.	sudo ldconfig
git [pkg]	Distributed version control: clone, add, commit, push, pull, status, log, diff, branch, merge, rebase.	git status
make / gcc / g++ / gdb [pkg]	Build automation, C/C++ compilers and the debugger from the standard development toolchain.	make -j\$(nproc)

23. Terminal Keyboard Shortcuts

Readline (the line-editing library used by Bash) and the terminal driver interpret these control keys before the command runs.

Shortcut	Action
Ctrl + C	Interrupt (SIGINT) the running foreground command.
Ctrl + Z	Suspend (SIGTSTP) the foreground command; resume with bg or fg.
Ctrl + D	Send end-of-file on an empty line, which exits the shell or ends input.
Ctrl + L	Clear the screen without deleting the current line.
Ctrl + A / E	Move the cursor to the beginning / end of the line.
Alt + B / F	Move one word backward / forward.
Ctrl + B / F	Move one character backward / forward.
Ctrl + U / K	Delete from the cursor to the beginning / end of the line.
Ctrl + W	Delete the word before the cursor.
Ctrl + Y	Yank back (paste) text previously cut with Ctrl+U/K/W.

Alt + .	Insert the last argument of the previous command.
Ctrl + R	Reverse-incremental search through command history; repeat to go further back.
Ctrl + G	Cancel the current history search.
Ctrl + P / N	Previous / next command in history (equivalent to the arrow keys).
Ctrl + T / Alt + T	Transpose the last two characters / words.
Ctrl + X then Ctrl + E	Open the current command in \$EDITOR for multi-line editing.
Ctrl + S / Q	Pause / resume terminal output flow control.
Tab / Tab Tab	Complete the current word; press twice to list all possible completions.
!! / !\$ / !n	History expansion: repeat the last command; its last argument; command number n.
^old^new	Re-run the previous command with old replaced by new.

24. Common Environment Variables and Shell Parameters

24.1 Environment Variables

Variable	Meaning
HOME	Current user's home directory.
PATH	Colon-separated directories searched for executable commands.
USER / LOGNAME	The current login name.
SHELL	Absolute path of the login shell.
TERM	Terminal type, which drives capabilities and colors.
PWD / OLDPWD	Current and previous working directories.
HOSTNAME	The machine's network hostname.
LANG / LC_ALL / LC_*	Locale: language, character set, collation and formatting.
EDITOR / VISUAL	Default editor launched by tools such as crontab, git and sudoedit.
PAGER / MANPAGER	Default pager for long output and manual pages.

PS1 / PS2	Primary and continuation prompt strings.
HISTSIZE / HISTFILESIZE / HISTCONTROL	How many commands history keeps in memory and on disk, and rules such as ignorespace/ignoredups.
IFS	Input field separator characters used when words are split.
DISPLAY	X11 display target for graphical programs.

24.2 Special Shell Parameters

Parameter	Meaning
\$0	Name of the shell or script.
\$1 ... \$9, \${10}	Positional arguments passed to a script or function.
\$#	Number of positional arguments.
\$@ / \$*	All positional arguments; "\$@" preserves each as a separate word.
\$?	Exit status of the most recent command (0 means success).
\$\$	PID of the current shell.
\$_	PID of the most recently started background process.
\$-	Currently enabled shell option flags.

25. How to List Every Command on Your System

Because a command is simply any executable on PATH, the authoritative list is generated on the machine itself. The following commands together enumerate external programs, shell built-ins, aliases and functions.

Enumerating the complete command set

```

1  # 1. Bash: every command name it can resolve (external + builtin + alias +
    function)
2  compgen -c | sort -u
3
4  # Count how many that is
5  compgen -c | sort -u | wc -l
6
7  # Only external executables found through PATH
```

```

8  compgen -c | while read -r c; do type -t "$c" | grep -q file && echo "$c";
   done | sort -u
9
10 # 2. Show the directories that make up PATH, one per line
11 printenv PATH | tr ':' '\n'
12
13 # 3. List every executable in the standard binary directories
14 ls -l /bin /sbin /usr/bin /usr/sbin /usr/local/bin /usr/local/sbin 2>/dev/null
   | sort -u
15
16 # 4. Find every executable file anywhere under PATH
17 IFS=':'; for d in $PATH; do find "$d" -maxdepth 1 -type f -executable
   2>/dev/null; done | sort -u
18
19 # 5. List Bash built-ins and keywords only
20 compgen -b; compgen -k
21
22 # 6. Explain a specific name (path, alias, builtin or function)
23 type -a ls
24
25 # 7. Which installed package provides a command (Debian / RHEL)
26 dpkg -S "$(which ls)"
27 rpm -qf "$(which ls)"

```

Manual sections 1 and 8 catalogue user and administration commands with documentation: browse them under `/usr/share/man/man1` and `/usr/share/man/man8`, or search them with `man -k ..`

26. One-Page Recipe Collection

Short, copy-ready combinations of the commands above for recurring tasks.

Frequently used one-liners

```

1  # --- Navigation and files ---
2  mkdir -p project/{src,docs,tests}           # create a folder skeleton
3  cp -av file file.bak                        # verbatim backup
4  find . -type f -empty -delete                # remove empty files
5
6  # --- Reading and filtering text ---
7  grep -RIn --exclude-dir=.git 'TODO' .       # find TODOs with line numbers
8  awk -F: '{s+= $3} END {print s}' file        # sum the third column
9  sort data | uniq -c | sort -nr | head        # frequency ranking
10 tail -F app.log | grep --line-buffered ERR  # live error watch
11 sed -n '10,20p' file                         # print lines 10 to 20
12

```

```

13  # --- Processes ---
14  ps -eo pid,ppid,pcpu,pmem,comm --sort=-pcpu | head    # top CPU users
15  sudo lsof -iTCP -sTCP:LISTEN -P -n                    # processes listening on ports
16  kill -9 $(pgrep -f runaway.sh)                        # force-kill by command match
17
18  # --- System and disk ---
19  du -sh * | sort -h                                     # size of each item, sorted
20  df -hT; free -h; uptime                               # quick health snapshot
21  journalctl --since '1 hour ago' -p err                # recent errors only
22
23  # --- Network ---
24  ss -tulpn                                              # listening sockets
25  dig +short myip.opendns.com @resolver1.opendns.com    # public IP
26  curl -I https://example.com                          # response headers only
27  rsync -avzP --delete src/ host:dest/                  # mirror a directory over SSH
28
29  # --- Archives ---
30  tar czf backup-$(date +%F).tar.gz data/              # dated compressed backup
31  tar xf archive.tar.*                                  # extract (auto-detects format)
32
33  # --- Safety patterns ---
34  set -euo pipefail                                     # strict mode at the top of scripts
35  !!:gs/old/new                                         # rerun last command, replace text
36  ^command^correctcommand                             # quick correction of last command

```



Learning path. Start with Sections 2-5 (help, syntax, files, viewing). Add Sections 6-7 (text processing and search) once you can navigate confidently. Sections 8-17 turn you into an operator; Sections 18 and 23-24 repay study forever because built-ins, shortcuts and variables work identically on every GNU/Linux machine you will ever touch.

(注：部分内容可能由 AI 生成)